

08 Pinout and packaging: orbit_demo, sky130hd layout variant sep

Package section 08_pinout_packaging of orbit_demo_design_review, written 2026-10-06 from the package files and the repository at commit 0495cfa. Links are relative to this file. Paths written as build/..., viz/..., site/... or docs/research/... are repository paths that are not copied into the package. Discrepancies are listed in the discrepancy register (09_review_notes).

Status labels: **VERIFIED** = a tool log or report shows it. **REPRODUCED 2026-10-06** = re-run for this package. **TARGET** = design goal, not measured. **ASSUMPTION. CLAIM (unverified)** = stated in a document, no log found. **MISSING. N/A** = not applicable (reason given). orbit_demo is a core-only standard-cell block. It has 215 signal pins on the die boundary, two supply pins formed by the met5 power straps, and no pad ring, IO cells, ESD structures, seal ring or package. This section documents the pins that exist and lists, as gaps, what a packaged chip would still need.

Design revision covered by this package

Item	Revision / identifier	Note
Design	orbit_demo (ORBIT-AI 4-lane INT8 digital demonstrator)	top module orbit_demo, LANES=4
RTL	rtl/orbit_demo.v, orbit_mac_lane.v, orbit_thermal_tmr.v, orbit_keep_reg.v at git commit 26566f0 (2026-09-29 18:54 UTC); unchanged through the package commit	sha256 a3fffb22... / efef3d33... / bba77a5a... / ea05bdc1... (full hashes in MANIFEST.sha256 at the package root)
Specification	docs/SPEC.md at 26566f0 (never revised)	sha256 aef5d9fb...
Concept brief	docs/orbit-ai-design-brief.pdf, "ORBIT-AI v0.1, 29 September 2026" (never revised; describes an earlier project state, see discrepancy register)	sha256 9d5f33d3...
Reviewed layout	ORFS sky130hd, variant sep (copy-separation fences), run 2026-09-29 21:56-22:12 UTC, outputs build/pd_sep/results/sky130hd/orbit_demo/sep/	6_final.gds sha256 cd19afa9...; 6_final.def f5c544f3...; 6_final.v 68093c41...; 6_final.spf 014e655a...
Layout configuration	pd/sky130hd_sep/config.mk and regions.tcl at c10c59b; constraint.sdc at 21917cf (set clk_period 7.2)	The run (21:56 UTC) used config.mk/regions.tcl as in ed553f8, identical to c10c59b once comments are stripped. It used the working-tree constraint.sdc with period 7.2 ns, which was committed in 21917cf at 22:00 UTC and is byte-identical to the flow copy build/pd_sep/sdc/sky130hd/sep.sdc. ed553f8's constraint.sdc has a 7.0 ns period, so a rebuild must not use it
Flow / tools	ORFS docker image tag openroad/orfs:latest; the only local image is sha256:2e5bf6fe865e... (created 2026-09-29 02:00 UTC, before the run). Linking the run to this digest is an ASSUMPTION: the run logged no digest. OpenROAD prints version "unknown". KLayout 0.30.12 (DRC/LVS). Yosys 0.69+154 (local sims/schematics)	No PDK commit is recorded in the logs
Process / library	SkyWater SKY130, sky130_fd_sc_hd, liberty sky130_fd_sc_hd_tt_025C_1v80 (the single corner ORFS optimised at)	—
Package assembled	2026-10-06; design inputs copied from the repository tree at commit 0495cfa (branch claude/hopeful-rubin-0yf8io)	Every later commit on the branch changes only review/ or the repository landing page README.md (added after packaging; not a design input), or is empty (git diff --name-only 0495cfa..HEAD outside review/ lists only README.md). Re-runs made for this package are dated 2026-10-06

Other implementations in the repo (sky130hd baseline at 7.0 ns, variant base; IHP SG13G2) are reference runs only and are NOT the reviewed layout.

Files in this folder

File	Content	Produced by	Status
pinout.csv	217 rows + header: pin, direction, use, layer, x_um, y_um, edge, spec_port, bit, spec_meaning	parse_pins.py on the final DEF	REPRODUCED 2026-10-06: re-generated from build/pd_sep/results/sky130hd/orbit_demo/sep/6_final.def, byte-identical; parser reports issues: []
parse_pins.py	DEF PINS parser, edge assignment, SPEC §2 cross-check (names, widths, directions)	—	—
rtl_ports.json	Yosys JSON of the elaborated RTL (read_verilog rtl/*.v; hierarchy -top orbit_demo; proc; write_json)	Yosys 0.69+154 (git sha1 30d62572e-dirty)	REPRODUCED 2026-10-06: re-generated, byte-identical
pin_map.png	Pin-position plot (Figure P1)	plot_pin_map.py, matplotlib 3.11.2, Python 3.11.15 (review venv)	plotted for this package from package files only
plot_pin_map.py	pin-map plotting script (reads pinout.csv and 6_final.def.gz)	—	—
pinout_packaging.pdf	this document rendered to PDF	md2pdf.py	—

Source layout for all pin data: 6_final.def.gz, which decompresses byte-identical to build/pd_sep/results/sky130hd/orbit_demo/sep/6_final.def (sha256 f5c544f37701038b0cf24fdc09481c58b249c425539b229d000ccb8a9df20ec1, written 2026-09-29 22:09:33 UTC). DEF 5.8, UNITS DISTANCE MICRONS 1000, DIEAREA (0 0) (346295 346295) = 346.295 x 346.295 um (decompressed lines 5-6). Conditions for every DEF-derived row below: ORFS sky130hd, variant sep, OpenROAD "unknown" in image openroad/orfs:latest (digest sha256:2e5bf6fe865e... is an ASSUMPTION; the run logged no digest), IO placement at the floorplan stage (3_2_place_iop).

Pin count check: DEF pins against RTL ports

Source	Ports	Signal bits	Supply terminals	Evidence	Status
RTL orbit_demo, LANES=4	18	215 (80 in incl. clk, 135 out)	none	rtl_ports.json ; orbit_demo.v lines 25-52	REPRODUCED 2026-10-06 (Yosys 0.69+154)
SPEC §2 port table	18	215 (8*LANES, 32*LANES with LANES=4)	not listed	SPEC.md lines 25-45	REPRODUCED 2026-10-06 (parse_pins.py: 0 missing, 0 extra, 0 direction mismatches)
Routed netlist 6_final.v	18	215	none	6_final.v.gz module header	REPRODUCED 2026-10-06 (port widths summed)
LVS reference 6_final.cdl	—	215	VDD, VSS	6_final.cdl.gz line 5: .SUBCKT orbit_demo with 217 terminals	REPRODUCED 2026-10-06
Routed DEF	217 pins	215 USE SIGNAL (80 INPUT, 135 OUTPUT), all PLACED, orientation N	VDD (USE POWER), VSS (USE GROUND), both DIRECTION INOUT, + SPECIAL, FIXED	6_final.def.gz , decompressed line 18346 PINS 217 ; to 19238 END PINS	REPRODUCED 2026-10-06
ORFS metrics	—	synth_design_io 215, floorplan_design_io 215	finish_design_io 217 (VDD/VSS pins are added by the PDN step)	1_synth.json line 2; 6_report.json line 50; results.md line 52	VERIFIED

Result: 217 DEF pins = 215 signal bits + VDD + VSS. Every one of the 215 signal pins has the RTL port name, bit index and direction. No test, scan or debug port exists. VDD/VSS are not in the SPEC or RTL port lists and appear only from the PDN step onward.

Pin function table (grouped by port)

All signal pins are block-boundary routing pins: met3 0.8 x 0.3 um on the W/E edges, met2 0.14 x 0.485 um on the N/S edges (DEF + LAYER met3 (-400 -150) (400 150) and met2 (-70 -242) (70 243)). They are not bond pads. Edge coordinates: W x = 0.400 um, E x = 345.895 um, S y = 0.242 um, N y = 346.052 um. Status of the whole table: REPRODUCED 2026-10-06 (from [pinout.csv](#), which matches the DEF; functions quoted from [SPEC.md](#) §2).

Port	Dir	Bits	Edge (pins)	Layer (pins)	Position along edge (um)	Function (SPEC §2)
clk	in	1	W	met3	y 334.900	Clock
rst_n	in	1	W	met3	y 310.420	Synchronous reset, active low
in_valid	in	1	W	met3	y 190.740	Input beat valid
in_ready	out	1	W	met3	y 189.380	Input beat accepted this cycle if in_valid is also high
in_first	in	1	S	met2	x 192.970	This beat starts a new sum
in_last	in	1	W	met3	y 192.100	After this beat, copy every lane's sum to the output buffer
in_a[31:0]	in	32	S 8, N 10, W 6, E 8	met2 18, met3 14	see lane table	Signed INT8 operands; lane i at [8*i +: 8]
in_b[31:0]	in	32	S 9, N 10, W 6, E 7	met2 19, met3 13	see lane table	Signed INT8 operands; lane i at [8*i +: 8]
out_valid	out	1	W	met3	y 182.580	Output buffer holds a presentable result
out_ready	in	1	W	met3	y 185.300	Consumer takes the result this cycle if out_valid is high
out_data[127:0]	out	128	W 82, S 24, N 22	met3 82, met2 46	see lane table	Result; lane i at [32*i +: 32], signed INT32
temp_valid	in	1	E	met3	y 171.700	Thermal reading is valid
temp_c[7:0]	in	8	E 8	met3 8	y 163.540-174.420 (bit order not monotonic)	Signed whole degrees Celsius
clear_fault	in	1	E	met3	y 196.180	Synchronous: clear fault, zero all lane storage, empty output buffer
fault	out	1	W	met3	y 196.180	Sticky: a duplicated copy disagreed
therm_state[1:0]	out	2	E 2	met3 2	y 177.140 ([0]), 181.220 ([1])	Voted thermal state: 0 NORMAL, 1 THROTTLE, 2 STOP, 3 unused (behaves as STOP)
therm_repair	out	1	E	met3	y 162.180	The three thermal copies do not all agree this cycle
shutdown_req	out	1	E	met3	y 175.780	therm_state[1]: STOP (or the unused code 3)
VDD	inout (power)	—	none: 12 met5 straps over the core	met5	straps x 2.30-344.08, centres y 29.92-329.12, pitch 27.2, width 1.6	Core supply (not in SPEC)
VSS	inout (ground)	—	none: 13 met5 straps over the core	met5	straps x 2.30-344.08, centres y 16.32-342.72, pitch 27.2, width 1.6	Core ground (not in SPEC)

fault (W edge) and clear_fault (E edge) sit on opposite die edges at the same y = 196.180 um.

Bus split by lane

Bus, lane	Bits	Edge, layer: bits (coordinate range, um)
in_a lane 0	[7:0]	S met2: all 8 (x 128.57-187.45)
in_a lane 1	[15:8]	N met2: all 8 (x 159.85-200.33)
in_a lane 2	[23:16]	W met3: [23:19], [16] (y 202.98-219.30); N met2: [18:17] (x 204.93-205.85)
in_a lane 3	[31:24]	E met3: all 8 (y 105.06-114.58)
in_b lane 0	[7:0]	S met2: all 8 (x 127.65-192.05)
in_b lane 1	[15:8]	N met2: all 8 (x 158.93-201.25)
in_b lane 2	[23:16]	W met3: [23:19], [16] (y 200.26-211.14); N met2: [18:17] (x 203.09-206.77)

Bus, lane	Bits	Edge, layer: bits (coordinate range, um)
in_b lane 3	[31:24]	E met3: [31], [29:24] (y 103.70-129.54); S met2: [30] (x 191.13)
out_data lane 0	[31:0]	S met2: 22 bits (x 10.81-62.33); W met3: [31:27], [25:23], [15], [5] (y 35.70-168.98)
out_data lane 1	[63:32]	N met2: 22 bits (x 114.77-153.41); W met3: [63:61], [58:57], [54], [45], [41:40], [38] (y 262.82-306.34)
out_data lane 2	[95:64]	W met3: all 32 (y 154.02-251.94)
out_data lane 3	[127:96]	W met3: 30 bits (y 57.46-175.78); S met2: [104], [96] (x 47.61-61.41)

Within each bus the bit order along an edge is not monotonic. Status: REPRODUCED 2026-10-06 (computed from [pinout.csv](#)).

Edge totals and pitch

Edge	Coordinate (um)	Layer	Pins	Span along edge (um)	Minimum centre spacing	Status
W	x = 0.400	met3	102	y 35.70-334.90	1.36 um (2 met3 tracks of 0.68 um)	REPRODUCED 2026-10-06
S	y = 0.242	met2	42	x 10.81-192.97	0.92 um (2 met2 tracks of 0.46 um)	REPRODUCED 2026-10-06
N	y = 346.052	met2	42	x 114.77-206.77	0.92 um	REPRODUCED 2026-10-06
E	x = 345.895	met3	29	y 103.70-196.18	1.36 um	REPRODUCED 2026-10-06
none	met5 straps over the core	met5	2 (VDD, VSS)	x 2.30-344.08	27.2 um strap pitch per net	REPRODUCED 2026-10-06

Track pitches from the DEF TRACKS statements (decompressed lines 136-139: met2 STEP 460, met3 STEP 680). The 2-track minimum spacing matches PLACE_PINS_ARGS = -min_distance 2 -min_distance_in_tracks ([config.mk](#) line 40).

Full CSV reference

[pinout.csv](#) lists every pin: x_um/y_um is the DEF pin origin (the pin-shape centre for signal pins, the origin of the first strap for VDD/VSS), edge is the nearest die boundary, and spec_meaning quotes SPEC \$2 with lane/bit and sign-bit annotations for in_a, in_b, out_data and temp_c. Regenerate with:

```
cd /home/user/chip
python3 -I review/orbit_demo_design_review/08_pinout_packaging/parse_pins.py \
    build/pd_sep/results/sky130hd/orbit_demo/sep/6_final.def pinout.csv
# expected: declared 217, parsed 217, issues [], by_edge W 102 / S 42 / N 42 / E 29 / met5 straps 2,
#           by_layer met3 131 / met2 84 / met5 2
```

Pin-position plot

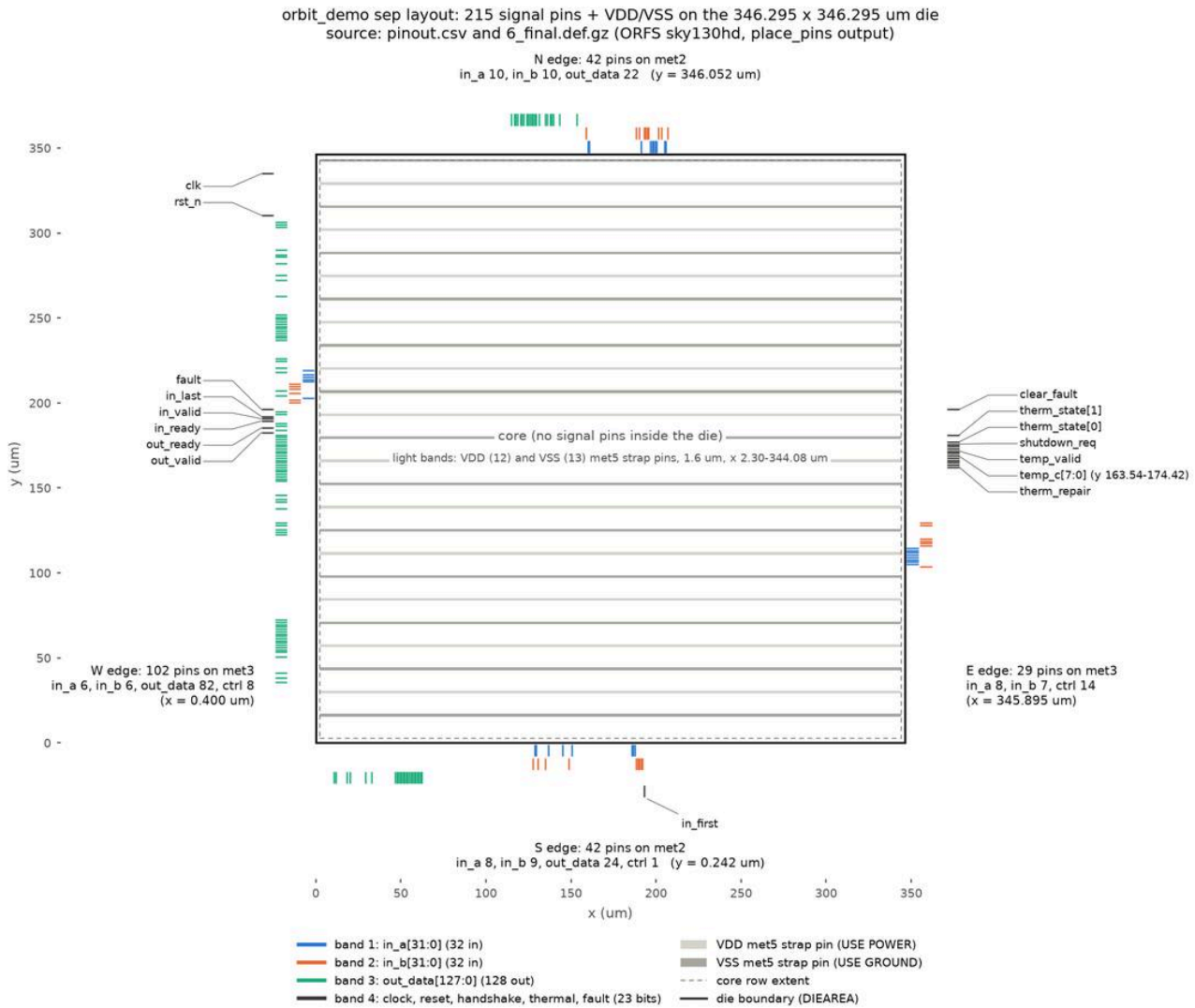


Figure P1. Source: `plot_pin_map.py` reading `pinout.csv` (215 signal pin origins, edges, ports) and `6_final.def.gz` (DIEAREA, ROW extent, the 12 VDD and 13 VSS met5 strap rectangles from the PINS section). The position of each tick along the edge is the real DEF pin origin. The offset band (1 = in_a, 2 = in_b, 3 = out_data, 4 = control/status) only separates the port groups and has no physical meaning. All signal pin origins lie within 0.4 um of the die boundary. Status: REPRODUCED 2026-10-06.

Pin assignment method

Item	Finding	Evidence	Status
Placement command	<code>place_pins -hor_layers met3 -ver_layers met2 -min_distance 2 -min_distance_in_tracks; 1254 available slots, 215 I/O, all with sinks, I/O nets HPWL 34567.11 um</code>	3_2_place_iop.log lines 10-18	VERIFIED
Order	<code>global_placement -skip_io</code> ran before <code>place_pins</code> , so pin positions follow the fenced cell placement	3_1_place_gp_skip_io.log line 12	VERIFIED
Pin constraints	none: no <code>set_io_pin_constraint</code> , <code>place_pin</code> or pin file in <code>pd/</code> , <code>scripts/</code> , <code>mk/</code>	repository search 2026-10-06	REPRODUCED 2026-10-06
Stability after IO placement	all 215 signal pin layers and locations in the final DEF equal <code>build/pd_sep/results/sky130hd/orbit_demo/sep/3_2_place_iop.tcl</code> (215 <code>place_pin ... -force_to_die_boundary</code> commands)	build file, not copied	REPRODUCED 2026-10-06
Comparison with the baseline layout	same <code>PLACE_PINS_ARGS</code> , different result: baseline DEF edges W 48, E 67, S 53, N 47	<code>parse_pins.py</code> on <code>build/pd/results/sky130hd/orbit_demo/base/6_final.def</code>	REPRODUCED 2026-10-06
Specified pinout	none exists	—	MISSING. Needed: a pin assignment agreed with the next level (die top or package), written as pin constraints and re-run through the sep flow

The pinout is tool output and will change with any placement change.

Electrical limits

No electrical specification exists for any pin. The values that do exist are standard-cell library limits on internal cells, or tool defaults. They are not pin ratings.

Item	What the files contain	Source	Status
Absolute maximum ratings (supply, pin voltage, current, storage temperature)	nothing	keyword scan of the repository; SPEC.md has no electrical section	MISSING
Recommended operating conditions (VDD range, junction temperature)	nothing specified. The only voltage is 1.80 V: the liberty corner <code>tt_025C_1v80</code> (operating_conditions voltage 1.8, temperature 25) and the ORFS default used by PSM ("Supply voltage : 1.80e+00 V")	6_report.log lines 6, 32; build/pd_sep/platform/cells.lib lines 31-34	VDD value: ASSUMPTION (tool default); specification MISSING
ESD rating (HBM/CDM), latch-up	no ESD structures in the layout; no rating	<code>padcells 0</code> (see Pad ring)	MISSING (no ESD exists; N/A to rate a bare core block)
Input/output DC levels (VIH, VIL, VOH, VOL), pin leakage, pin capacitance	nothing	—	MISSING
Output drive	not specified. Each of the 135 outputs is driven directly by one <code>sky130_fd_sc_hd__clkdlbbuf4s50_1</code> (X pin); its liberty limits at TT are <code>max_capacitance 0.156565 pF</code> and <code>max_transition 1.495315 ns</code> . The SDC has no <code>set_load</code> , so no external load is checked against them	DEF NETS (e.g. decompressed line 93669 <code>in_ready (PIN in_ready) (output87 X);</code> build/pd_sep/platform/cells.lib lines 47714-47715	Cell limits VERIFIED; as a pin drive rating MISSING
Input load	not specified. Each input net connects to exactly one buffer A pin (9 cell types, liberty pin capacitance 0.001727 pF for <code>buf_2</code> to 0.009187 pF for <code>buf_12</code> ; <code>clk</code> drives <code>clkbuf_0_clk</code> , a <code>clkbuf_16</code>). 8 input nets (<code>in_a[23]</code> , <code>in_a[25]</code> , <code>in_a[28]</code> , <code>in_b[16]</code> , <code>in_b[25]</code> , <code>temp_c[7:5]</code>) also carry an antenna-repair <code>sky130_fd_sc_hd__diode_2</code> ; these are not ESD protection	DEF NETS section; build/pd_sep/platform/cells.lib	Cell values VERIFIED; as a pin specification MISSING
Input transition	liberty default <code>max_transition : 1.5 ns</code> and input-pin <code>max_transition 1.5 ns</code> are library design rules. The SDC has no <code>set_input_transition</code> , so STA assumes 0.00 ns slew at every input port (e.g. <code>in_a[27] (in)</code> slew 0.00)	build/pd_sep/platform/cells.lib line 26; worst_setup_path.txt line 12	Library limit VERIFIED; input-slew requirement MISSING
Max slew / cap / fanout checks	PASS at TT on internal nets; at <code>ss_100C_1v60</code> , 37 max-slew and 12 max-capacitance violating pins	results.md line 17; sta_corners_sep_supplement.txt	VERIFIED (TT); REPRODUCED 2026-10-06 (corners)
Supply current	52.1092 mW total at default switching activity (OpenSTA <code>report_power</code> , TT, no VCD/SAIF), about 28.9 mA average at 1.80 V (arithmetic, not a tool output)	power_default_activity.txt line 14; results.md	Power VERIFIED (estimate); current rating MISSING
Brief figures (16 tiles, 200-800 MHz, 40 W)	figures for the unbuilt ORBIT-AI chip: 16 tiles = proposed concept architecture (p.1-2); 200-800 MHz = clock target (p.1 "200-800 MHz target"); 40 W = sizing assumption (p.1 "40 W chip sizing assumption - not an estimate", p.2 "for conceptual sizing only. This is not a power estimate")	brief p.1-2	200-800 MHz TARGET; 40 W ASSUMPTION; none applies to this block (N/A)

Package

Item	Finding	Source	Status
Package for this block	none defined; the block is a core-only layout	design data (no package file of any kind)	N/A: <code>orbit_demo</code> is a core block, not a chip
Package for a fabricated chip	not selected. Brief p.2: "Mission orbit, lifetime, manufacturing process, package and radiation acceptance levels remain undecided." p.4, next step 2: "Choose characterized process/library, memory, package and interfaces". p.1: "No physical chip fabricated."	brief p.1, 2, 4	MISSING
3D viewer "Packaged chip" tab	a procedural mockup. viz/index.html line 519: "Mockup. The package, leadframe, bond wires, pad ring and die outline are drawn procedurally to show how this die would be packaged; none of them is in the routed data." Line 357: "Mockup · package and pad ring are not in the routed data". Line 556: "Body, lead and pad sizes follow typical QFN/QFP practice, not a datasheet." Line 1953: "QFN/QFP-style package (a MOCKUP: nothing of it is in the routed data)". Parts list lines 2237-2242 mark lid, bond wires, die attach, leadframe and leads "illustrative" and the die "core measured, ring illustrative". Mockup constants: <code>PAD_W = 80</code> , <code>PAD_PITCH = 100</code> , <code>LEAD_PITCH = 500</code> , <code>DIE_T = 280 (um)</code> ; default ring "5 per side: a QFN-20" (line 2023)	viz/index.html (repository, not copied)	N/A: illustration, not package data
Public site package scenes	site/chip-space/index.html line 640 "Mockup package · not to scale", line 1194 "MOCKUP PACKAGE", 40 leads and 40 pads (lines 1148-1149); site/chip-story/index.html line 188 "The package is an illustrative mockup. The die geometry is measured."	repository, not copied	N/A: illustration. The mockups disagree with each other (QFN-20, QFP-style, 40-lead); see discrepancy register (09_review_notes)

No mockup image is embedded here because none is package evidence.

Pad ring and IO cells

Item	Finding	Source	Status
Pad cells	<code>*_instance__count__padcells 0</code> at every ORFS stage from synthesis to finish	1_synth.json line 12; 6_report.json line 60	VERIFIED
Pad power group	<code>Pad 0.00e+00 ... 0.0%</code>	power_default_activity.txt line 12	VERIFIED

Item	Finding	Source	Status
Cell library content	all 18175 DEF COMPONENTS are sky130_fd_sc_hd_* cells; the GDS top cell orbit_demo references 153 cells, none of them IO, pad or seal-ring cells	6_final.def.gz COMPONENTS 18175 (decompressed line 169); 6_final.gds.gz read with KLayout 0.30.12 Python	REPRODUCED 2026-10-06
Die edge	GDS top-cell bounding box (0, 0; 346.295, 346.295) um = DIEAREA: no seal ring or scribe structure outside the core margin	6_final.gds.gz, KLayout 0.30.12	REPRODUCED 2026-10-06
Core power ring	none: SPECIALNETS contain only STRIPE, FOLLOWPIN and DRCFILL shapes (met1 followpins, 1.6 um met4 and met5 stripes)	6_final.def.gz, decompressed lines 19239-27606	REPRODUCED 2026-10-06
Pad ring	none	—	N/A for this core block; MISSING for a chip

Required external connections

These requirements follow from the RTL, the SPEC and the SDC. None is written down as an interface specification. Pin locations are in the pin function table.

Connection	Pins	Requirement as far as the design data support it	Source	Status
Clock source	clk (W, met3, 0.400/334.900 um)	One posedge clock domain; no oscillator, PLL or clock pad on chip. The SDC defines create_clock -name clk -period 7.2000 with set_propagated_clock; no uncertainty, jitter, source latency or duty-cycle constraint, and the clock arrives ideal (clk (in) slew 0.00). The pin drives clkbuf_0_clk (sky130_fd_sc_hd_clkbuf_16) directly. At 7.2 ns, timing closes at tt_025C_1v80 (setup WNS 0.031 ns, hold WNS 0.436 ns) and at ff_n40C_1v95 (setup WNS +2.276 ns, hold WNS +0.281 ns), and fails at ss_100C_1v60 (setup WNS -6.131 ns, 516 failing endpoints, min period 13.33 ns). ORFS optimised at TT only	6_final.sdc lines 8-9; worst_hold_path.txt lines 11-12; DEF net clk (decompressed line 67767); results.md lines 15-16; sta_corners_sep.txt lines 3-4, 13-15; sta_corners_sep_supplement.txt lines 30, 42	TT timing VERIFIED; corners REPRODUCED 2026-10-06 (OpenSTA in the ORFS image, nominal OpenRCX SPEF); clock source spec (frequency range, jitter, duty cycle) MISSING
Reset	rst_n (W, met3, 0.400/310.420 um)	Synchronous, active low, highest priority in every flip-flop (if (!rst_n) first in every always @(posedge clk)). The clock must run while rst_n is low. One rising clk edge with rst_n low clears fault_q and out_valid_q, zeroes all 16 lane storage registers and puts the three thermal copies in STOP with phase 0: formally proven from an arbitrary start state (clear task, properties P8_any_*, P5_any_reset_stop, k-induction k = 3). No reset synchronizer or power-on reset; rst_n is timed as a data input with 1.44 ns input delay, so its deassertion must be synchronous to clk. The directed testbench holds reset for 4 cycles	orbit_demo.v lines 107-110; orbit_keep_reg.v lines 26-31; orbit_thermal_tmr.v lines 12-13; formal_summary.md lines 32, 228-235; orbit_demo_fv.sv lines 563-585; tb_orbit_demo.v lines 488-502	Single-edge reset: REPRODUCED 2026-10-06 (SBY v0.69, Yices 2.7.0). Documented reset pulse width, power-up sequence and reset source: MISSING
Temperature reading	temp_valid, temp_c[7:0] (E, met3, y 163.54-174.42)	A digital value (signed whole degrees C) from an external source, timed like any input (1.44 ns input delay, synchronous to clk, no synchronizer). The physical sensor is not implemented ("Not included: ... physical sensor interfaces", RTL header; brief p.3 "Physical sensors, calibration and an independent shutdown circuit are not implemented"). After reset the design is in STOP and admits nothing until a valid reading at or below 70 C; !temp_valid or temp_c >= 95 forces STOP. Thresholds 80/95/70 C are RTL parameters, "illustrative prototype values, not device ratings"	orbit_demo.v lines 9-10; orbit_thermal_tmr.v lines 7-13; SPEC.md lines 100-124	Behaviour REPRODUCED 2026-10-06 (formal thermal task PASS); thresholds TARGET; sensor type, accuracy, update rate and synchronisation MISSING
Supply	VDD, VSS met5 strap pins	The only supply connection points are the 12 VDD and 13 VSS horizontal met5 straps (1.6 um wide, x 2.30-344.08 um, 27.2 um pitch). They connect through met4 stripes and met1 followpins to every cell's VPWR/VPB (VDD) and VGND/VNB (VSS). One voltage domain. There are no power pads or bond pads; the next level must contact these straps from above or route to them. Static IR drop at 1.80 V and default activity: VDD worst 0.454 mV, VSS worst 0.628 mV (0.03 %)	6_final.def.gz decompressed lines 18347-18376 (PINS), 19239-19240 and 23252 (SPECIALNETS); 6_report.log lines 29-49; re-run ir.log lines 5-42; tech LEF read by the run: 1_synth.log line 7 (limits read from the image copy on this host, ASSUMPTION same file)	Geometry VERIFIED; IR VERIFIED and REPRODUCED 2026-10-06 (OpenROAD PSM, no VCD/SAIF); PSM EM currents at default activity are in ir.log (maximum resistor current 4.05e-04 A VDD, 4.33e-04 A VSS; REPRODUCED 2026-10-06) but were not checked against the DCCURRENTDENSITY limits of the tech LEF sky130_fd_sc_hd.tlef (MISSING); supply voltage specification and per-strap current budget MISSING
Input stream (producer)	in_valid, in_first, in_last, in_a[31:0], in_b[31:0] in; in_ready out	Valid/ready: a beat transfers on a rising edge with in_valid && in_ready	SPEC §2-3; formal_summary.md lines 16-18	RTL and SPEC agree (manual reading); handshake properties P2-P4 REPRODUCED 2026-10-06 (formal handshake/handshake_pdr PASS)

Connection	Pins	Requirement as far as the design data support it	Source	Status
Output stream (consumer) and the out_ready path	out_valid, out_data[127:0] out; out_ready in	in_ready = admit & ~fault_q & ~mismatch & ~clear_fault & (~out_valid_q out_ready): out_ready and clear_fault reach the in_ready output combinational. The SDC times this as an input-to-output path with a 7.2 - 1.44 - 1.44 = 4.32 ns budget. Worst in-to-out setup endpoint is in_ready: slack +2.577 ns (TT), +1.195 ns (ss_100C_1v60), +3.165 ns (ff_n40C_1v95). The reg-to-out path to in_ready fails at ss_100C_1v60 (-4.142 ns). out_valid = out_valid_q & ~fault_q & ~mismatch can fall without out_ready on a storage mismatch, clear_fault or reset, so the consumer must accept a withdrawn result	orbit_demo.v lines 64-73; SPEC.md lines 66-78, 91-97; constraint.sdc lines 7-8; sta_corners_sep_supplement.txt ; sta_extra_tt.log lines 40-70; formal_summary.md lines 16-18	in_ready/out_valid equations: RTL read manually; P3 (out_valid held, out_data stable while !out_ready) REPRODUCED 2026-10-06 (formal handshake/handshake_pdr PASS); path slacks REPRODUCED 2026-10-06 (the -format end report does not name the start point)
Host control and status	clear_fault in; fault, therm_state[1:0], therm_repair, shutdown_req out	clear_fault is synchronous; in_ready is low while it is high. fault is sticky and registered. shutdown_req = therm_state[1] (RTL comment "request safe shutdown"); acting on it is outside the block (brief p.1 shows the "Power / current supervisor" as "External hardware to be selected")	SPEC.md lines 41-45, 95-98; orbit_demo.v lines 47-51, 125; formal_summary.md lines 18, 22	RTL and SPEC agree (manual reading); in_ready low while clear_fault (P4) and the clear_fault effect (P8) REPRODUCED 2026-10-06 (formal handshake/handshake_pdr, datapath, clear PASS); external supervisor MISSING
IO timing environment	all non-clock inputs, all outputs	set clk_io_pct 0.2 (20 % of the 7.2 ns period): set_input_delay 1.44 ns on the 79 non-clock inputs (including rst_n) and set_output_delay 1.44 ns on the 135 outputs, "for the (unknown) outside world". The Environment and Design Rules sections of 6_final.sdc are empty: no set_driving_cell, set_load, set_input_transition or set_clock_uncertainty	constraint.sdc lines 5-8, 19-27; 6_final.sdc (79 set_input_delay, 135 set_output_delay; Environment and Design Rules headers at lines 224-229 with no commands)	SDC content VERIFIED; as an interface budget ASSUMPTION; agreed IO budgets MISSING

The worst setup path of the whole design starts at an input port (in_a[27], E edge, 345.895/109.140 um) and ends at g_lane[3].u_lane.u_res_a/q[29] near the W edge with 0.031 ns slack ([worst_setup_path.txt](#)). The assumed 1.44 ns input delay is part of that path, so any change to the IO budget changes the timing result.

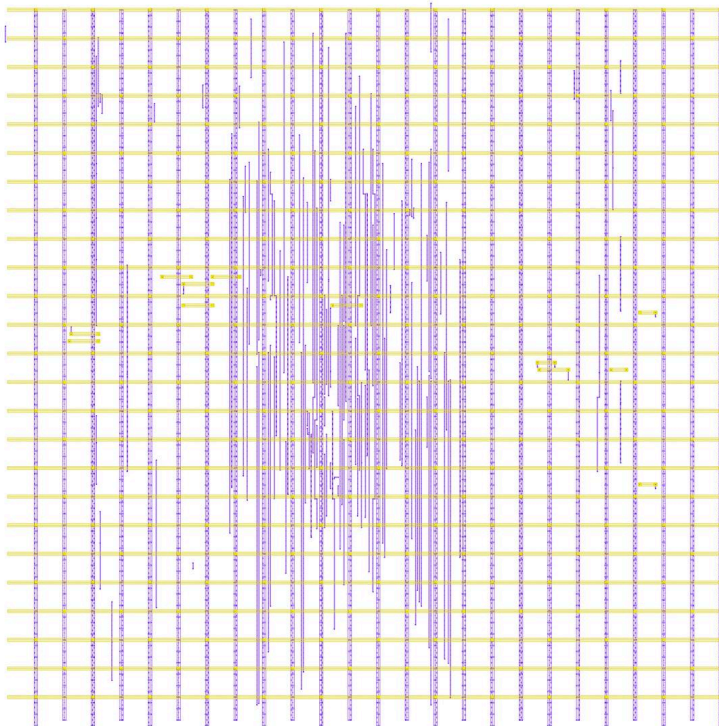


Figure P2. [v02_full_die_pdn_met4_met5.png](#), rendered from 6_final.gds with KLayout 0.30.12 by [render_gds.py](#) (section 06): met4 (vertical) and met5 (horizontal) straps over the full die, plus the signal wires routed on met4/met5. The horizontal met5 straps are the VDD/VSS pins. Status: REPRODUCED 2026-10-06 (render of 6_final.gds, see 06 section 6.1).

Gaps to a packageable die

Stated as missing items. None of them is designed or started in the repository.

Gap	Current state	What is needed	Status
Package selection	undecided (brief p.2)	package type, body and lead/ball count, thermal path; package drawing; die-to-lead or die-to-ball map	MISSING
IO budget	215 signal bits + supply on minimum-size routing pins	a pad count decision: either a pad-limited die with about 215 signal pads plus supply pads, or on-chip IO reduction (multiplexing or serialisation of in_a, in_b, out_data), which changes the RTL interface	MISSING
Pad ring with IO cells	padcells 0; no IO library read (only sky130_fd_sc_hd__tt_025C_1v80.lib)	IO/pad cells from a characterized IO library for SKY130, pad frame floorplan, corner and filler pads, IO cell timing in the SDC	MISSING

Gap	Current state	What is needed	Status
ESD and latch-up protection	none (the 20 diode_2 cells are antenna repair)	ESD clamps on every signal and supply pad, supply clamps, ESD/latch-up rules check	MISSING
Supply pads and distribution	VDD/VSS exist only as met5 straps over the core; no core ring	VDD/VSS pad count from the current budget, core power ring, connection from pads to the straps, EM check with real activity	MISSING
Seal ring and die edge	GDS top-cell bounding box equals DIEAREA (346.295 x 346.295 um); nothing outside the core margin	seal ring, scribe/edge structures as required by the foundry	MISSING
Clock and reset entry	clk and rst_n are plain block pins; no reset synchronizer, no power-on reset	clock input pad and clock specification (frequency, jitter, duty cycle, set_clock_uncertainty); reset pad and reset synchronizer or a documented external reset protocol	MISSING
Electrical specification	none (see Electrical limits)	absolute maximum ratings, recommended operating conditions, IO DC/AC characteristics from the chosen IO library	MISSING
Fixed pinout	placer output, changes with placement	pin constraints matching the pad ring order, re-run of the sep flow	MISSING
Chip-level timing	fails at ss_100C_1v60 (-6.131 ns at 7.2 ns); closes at TT and ff_n40C_1v95	multi-corner closure with pad delays, set_driving_cell/set_load from the board or package model	MISSING
Chip-level sign-off	block-level KLayout DRC with the ORFS deck (BEOL only, FEOL disabled) and KLayout LVS against the OpenROAD CDL; see 07_verification	full-chip DRC with a sign-off rule deck (FEOL, density, antenna), LVS including pads and seal ring	MISSING
Production test access	18 functional ports, no scan or test port; all 521 flip-flops are sky130_fd_sc_hd_dfxtpt_1 (no scan flip-flops in the DEF COMPONENTS)	a test strategy (scan insertion or functional test through the pads)	MISSING
Temperature sensor	external digital input only	on-chip sensor/ADC or a defined external sensor interface, with accuracy and latency requirements	MISSING

Discrepancies noted in this section

All are entered in the discrepancy register (09_review_notes); only a pointer is given here.

Topic	Evidence	Severity
docs/research/labelled-chip-models.md gives the pin edges N 47, S 53, E 67, W 48. These are the baseline layout's pins (the document names build/pd/.../base/6_final.def as its source, line 13), not the reviewed sep layout (W 102, S 42, N 42, E 29)	lines 66-72; parse_pins.py on both DEFs	medium (risk of using the wrong pinout); see discrepancy register (09_review_notes)
viz/layout.json lists VDD/VSS as 'dir': 'I', 'side': 'N', note "met5/met4 straps"; the DEF has DIRECTION INOUT, met5-only pin shapes spanning the core	viz/layout.json pins.list; DEF lines 18347, 18362	low; see discrepancy register (09_review_notes)
The viewer package tab labels the 346.295 um die as "Core (measured)"; the core rows span 341.780 x 340.000 um	viz/index.html lines 519, 2355; results.md lines 38-39	low; see discrepancy register (09_review_notes)
DEF pin clk is USE SIGNAL while net clk is USE CLOCK; SPEC calls the port "Clock"	DEF lines 18382, 67767; SPEC line 29	low; see discrepancy register (09_review_notes)
The package mockups disagree (QFN-20 default ring, QFP-style "All pins" ring, 40-lead site scene); none is a selected package	viz/index.html line 2023; site/chip-space/index.html lines 1148-1149	low; see discrepancy register (09_review_notes)
SPEC §4 and the SDC comment name only out_ready -> in_ready as the combinational path; the RTL equation also makes clear_fault -> in_ready combinational (SPEC §5 states only that in_ready is low while clear_fault is high). Both are timed by the in-to-out STA	orbit_demo.v lines 70-71; SPEC.md lines 78, 97; constraint.sdc lines 7-8	low (documentation); see discrepancy register (09_review_notes)
The public site states "139 MHz Clock, timing met" without naming the corner (139.49 MHz is the TT result); at ss_100C_1v60 the layout fails at 7.2 ns	site/chip-space/index.html lines 534, 663; sta_corners_sep.txt line 14	medium; see discrepancy register (09_review_notes)

Reproduction (2026-10-06)

```
cd /home/user/chip
# pinout table and SPEC cross-check (byte-identical to pinout.csv)
python3 -I review/orbit_demo_design_review/08_pinout_packaging/parse_pins.py \
    build/pd_sep/results/sky130hd/orbit_demo/sep/6_final.def /tmp/pinout.csv
# RTL port list (byte-identical to rtl_ports.json)
PATH=/opt/eda/oss-cad-suite/bin:$PATH yosys -q -p 'read_verilog rtl/orbit_demo.v rtl/orbit_mac_lane.v \
    rtl/orbit_thermal_tmr.v rtl/orbit_keep_reg.v; hierarchy -top orbit_demo; proc; write_json /tmp/rtl_ports.json'
# pin map (needs matplotlib)
python3 review/orbit_demo_design_review/08_pinout_packaging/plot_pin_map.py
```